

Hints to Selected Problems (3rd ed.)

We include here some very brief hints to some of the problems. Occasionally we include a full solution, but mostly we only include a suggestive word or two to spur on your investigations. For more complete solutions to most of the problems, consult the solutions manual. You can find more information about this at the Wiley web site for this book.

Chapter 2

2.1.20 Conservation of mass.

2.1.25 Here are the answers. If you did not get these answers, keep thinking about what tricked you. (a) 30. (b) 1. (c) 5. (d) 5. (e) All positive integers.

2.2.11 Any time a problem involves doubling, like this one, you should think about writing numbers in binary (base 2).

2.2.14 It's too hard to draw pictures for $n \geq 4$. Try to represent the situation 2-dimensionally. If mapmakers can do it, so can you. The answer, by the way, is not a power of 2. It is $n^2 - n + 2$.

2.2.22 The exponents are numbers whose base-3 representation only contains the digits 0 and 1. This looks like base-2 numbers!

2.2.25 Once again, think about base-2 representation.

2.2.30 (a) What if $n = 41$?

2.3.14 (a) You need the fundamental theorem of arithmetic (see Section 7.1).

2.3.22 If the conclusion were false, then there exist t_1, t_2, u_1, u_2 such that $t_1 t_2 \in U$ and $u_1 u_2 \in T$. Use these to get a contradiction by finding a product of several numbers which lies in both U and T .

2.3.23 It is helpful to think *dynamically*; imagine slowly drawing in the $(n + 1)$ st line and keep a running count of all the new areas that it creates.

2.3.27 Equivalently, you want to show that $7^n = 1 + 6N$, where N is an integer.

2.3.32 Some of these problems require strong induction. Another idea of “strengthening” is to try to prove *more* than what is asked (see Example 2.3.10 on page 48). For example, with problems (f) and/or (g), you may need to prove two different statements, not one, and use each statement to help the other.

2 HINTS TO CHAPTER 3

- 2.4.10** Draw a distance-versus-time graph.
- 2.4.12** Draw a distance-versus-time graph.
- 2.4.18** Whenever you see sums of squares, think about distance in a coordinate system [since the distance between (x_1, y_1) and (x_2, y_2) is $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$].
- 2.4.19** Three dimensions are too hard!
- 2.4.20** Think about right triangles. Convert this to a geometry problem.
- 2.4.21** The algebra is too hard. Try a picture. If the “floor” brackets confuse you at first, temporarily pretend that they aren’t there.
- 2.4.23** See the hint for 2.4.18.
- 2.4.27** The problem leads one to think about counting solutions to inequalities, and the solution points are lattice points on the plane.
- 2.4.29** This is a good “backburner” problem. Consult the sections on symmetry and invariants (Sections 3.1, 3.4, respectively) for ideas.

Chapter 3

- 3.1.13** You will need to use the reflection tool (Example 3.1.5 on page 63) *twice*.
- 3.1.19** After the bugs have turned, say, 1 degree, they are still at the vertices of a square, and they still haven’t crashed into one another. So they will turn another 1 degree, and then, ...
- 3.1.20** The center of the table is a “distinguished” point.
- 3.1.21** Virtually any problem involving actual reflection will benefit from the reflection tool (Example 3.1.5 on page 63). The strategy is to try to turn a jagged path into a nice straight line.
- 3.1.24** The temperature function $T(x, y, z)$, unfortunately, is not symmetric in the three variables. So *impose* symmetry: consider the new function

$$T(x, y, z) + T(y, z, x) + T(z, x, y).$$

Reread Example 3.1.7 on page 64 for inspiration.

- 3.1.25** See 3.1.24 above.
- 3.1.28** (c) Use metaphorical symmetry: the “dual” of a bridge you can cross is a bridge you have cut.
- 3.2.7** Look at a square which contains the smallest value.
- 3.2.11** Consider the two people whose distance apart is the smallest.
- 3.2.12** Run the movie backwards.
- 3.2.14** At “worst,” 1 and n^2 are diagonally opposite. In all other cases there is a shorter path connecting these two numbers.
- 3.2.15** (d) *Solution:* Let t be chosen so that $r := b - at$ is the smallest possible nonnegative value (t is a positive integer). We claim that $0 \leq r < a$. Assume not; since $r \geq 0$, then we would have $r \geq a$.

But then $r - a = b - a(t + 1)$ is nonnegative, yet smaller, contradicting the minimality of r . ■

3.3.11 There seem to be not enough pigeons, but you can bypass this problem by looking at cases: What if someone knows no one? What if everyone knows at least someone?

3.3.16 Penultimate step: look at *rectangles* of size 0.02.

3.3.18 Penultimate step: there are many ways to force numbers to be relatively prime. Think of some simple ways.

3.3.19 How “far” is each person from his correct entrée?

3.3.21 Think about the parity of exponents.

3.3.28 Think about diagonals and things that “act like” diagonals.

3.4.20 Parity provides much information.

3.4.18 Impose cartesian coordinates on the frogs; it adds free information!

3.4.22 Try some examples. Look at divisibility.

3.4.25 The pair of numbers a, b can be nicely represented as a point in the plane. Try drawing some very careful pictures to keep track of your experiments.

3.4.27 Reading from left to right, keep track of changes.

3.4.30 First show that every person’s weight must have the same parity.

3.4.31 Three dimensions are too hard. Make it easier!

Chapter 4

4.1.8 See Problem 3.3.11.

4.1.9 An equivalent problem: Prove that if you color the edges of a K_6 with two colors, then there will be a monochromatic triangle. To see this, start with an arbitrary vertex. By pigeonhole, at least 3 of the 5 edges emanating from this vertex are, say, blue. Then what?

4.1.11 Argue by contradiction and use pigeonhole-style thinking. Consider, for example, when $v = 12$. If the graph were not connected, then there would have to be a “clique” of with 6 or less vertices (a clique being a subset of vertices none of which are neighbors of any vertices outside the clique).

4.1.15 (a) Use extreme principle. Consider the the longest such “oriented path,” and show that it must include all players.

(b) The equivalent statement: a directed complete graph possesses a Hamilton path.

4.1.16 Penultimate step: can we recast the problem in such a way that an Eulerian path would solve it? Or might a Hamilton path do the trick? Could either work? Be flexible about which entities play the role of vertices or edges. For example, one interpretation makes each of the 28 dominos a different vertex. What does an edge mean, then? Another possibility is to make each of the 7 numbers 0, 1, 2, ..., 6 a vertex ...

4 HINTS TO CHAPTER 5

4.1.18 Analyze degree numbers; use handshake lemma.

4.1.19 Devise an algorithm, where you travel first on a 1-cube, then a 2-cube, etc. Prove that your algorithm works using induction.

4.2.9 (a) Use a picture to show that $z\bar{z} = |z|^2$.

(c) The four points $z, 1-z, -1, 0$ form a rhombus (a parallelogram with 4 equal side lengths). Recall (and prove!) that the diagonals of a rhombus are perpendicular and bisect one another.

(e) Verify that if $\operatorname{Re}(w/z) = 0$, then w and z are perpendicular as vectors.

(f) Take absolute values of both sides of the equation $(z-1)^{10} = z^{10}$.

4.2.20 If z, w lie on the unit circle, then the four points $0, z, w, z+w$ form a rhombus. See Problem 4.2.9(c) as well.

4.2.23 Show that two of the zeros of this polynomial are ω and ω^2 , where $\omega = e^{2\pi i/3}$ is a cube root of unity. This implies that a certain quadratic polynomial divides $z^5 + z + 1$.

4.2.25 Two approaches: One is to replace $\sin t$ with $(e^{it} - e^{-it})/2i$. The other is to use 4.2.9(c). You may need the factor theorem (see Section 5.4).

4.2.27 The polynomial $x^4 + x^3 + x^2 + x + 1$ is shouting at you, “5th roots of unity!”

4.2.33 Penultimate step: if the line segments are the vectors z, w , then $z = \pm iw$ is equivalent to saying that z, w are perpendicular and of equal length.

4.3.10 You will need the technique of partial fractions (see footnote on page 136).

4.3.12 Note that $1 + x + x^2 + x^3$ factors.

4.3.15 Use the geometric series tool.

4.3.16 Let a_n be the number of partitions of a positive integer n into parts that are not multiples of three and let b_n be the number of partitions of n in which there are at most two repeats, and let the generating functions for $(a_n), (b_n)$ be $A(x), B(x)$, respectively. Show that

$$A(x) = (1 + x + x^2 + \cdots)(1 + x^2 + x^4 + \cdots)(1 + x^4 + x^8 + \cdots)(1 + x^5 + x^{10} + \cdots) \cdots$$

and

$$B(x) = (1 + x + x^2)(1 + x^2 + x^4)(1 + x^3 + x^6)(1 + x^4 + x^8) \cdots$$

4.3.18 Use the geometric series tool. Think about dice.

4.3.19 Consider the polynomial $D(x) := x + x^2 + x^3 + x^4 + x^5 + x^6$. Now square it; what does this represent?

Chapter 5

5.1.9 Penultimate step: to show that $\lfloor u \rfloor = \lfloor v \rfloor$, show that $N \leq u, v < N + 1$ for some integer N . Note that $4n + 2$ is never a perfect square.

5.1.13 Many approaches work, including induction. There are a number of slick methods, but here is a brute-force approach that isn't too hard, but instructive: write n in binary (base 2).

5.2.26 It is helpful to know that all answers to AIME problems are integers between 0 and 999. Let the number in question be N . Notice that N must end in a 2, so we can write $N = 10a + 2$, where a is a one- or two-digit number.

5.2.27 Use 5.2.10.

5.2.34 The fact that $(16x^2 - 9) + (9x^2 - 16) = (25x^2 - 25)$ is *not* a coincidence.

5.3.12 Use the catalyst tool (see Example 5.3.3).

5.3.15 Note that if the sequence a_n is quadratic function of n , then the difference sequence $\Delta(a)_n := a_{n+1} - a_n$ will be a linear sequence, and *its* difference sequence $\Delta(\Delta(a))$ will be a constant sequence!

5.3.18 The harmonic series diverges.

5.3.19 Get an upper bound on the sum by estimating the size of the $1/k$ terms when k has a one-digit, two-digit, etc. Geometric series are easy to use, so play around with them.

5.4.9 Plug in the zeros of $x^3 - x$.

5.4.12 Start by letting $x = \sqrt{2} + \sqrt{5}$, and then do smart algebraic things until you have a polynomial with integral coefficients.

5.4.13 See Example 7.1.7 on page 235 for a solution.

5.4.14 Think about parity.

5.4.16 Plug in $x = a, b, c$. If it factors, then one factor is linear, the other is quadratic.

5.4.22 Note that the converse to Problem 5.4.8 is also true.

5.5.27 Rationalize the numerator.

5.5.28 Use AMGM.

5.5.29 Verify that if any term in the product is greater than 3, the product can be increased by breaking up this term into two terms, one of which is 2 or 3.

5.5.34 Think about coordinates on the plane.

5.5.46 Notice that Cauchy-Schwarz implies that if $r_1, r_2, \dots, r_n$ are real, then

$$n \sum r_i^2 \geq \left(\sum r_i \right)^2.$$

In addition, use the relationship between roots and coefficients (see page 175).

Chapter 6

6.1.20 (a) Consider the problem of counting the number of ways you can choose two people from a pool of n men and n women. What are the cases?

6.1.23 See page 205 for a solution.

6 HINTS TO CHAPTER 7

6.1.24 For (a), see page 205. For ideas about (b), look at Section 6.3.

6.1.26 Classify the color schemes according to symmetry. Different things will be overcounted by different amounts. Check your ideas with a much smaller case, such as a 3×3 board.

6.1.29 Here is the answer, but ponder it until you understand *why* it is true. The key tool used is the mississippi formula (6.1.10):

$$(x_1 + x_2 + \cdots + x_n)^r = \sum \frac{r!}{a_1! a_2! \cdots a_n!} x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n},$$

where the indices a_i range through all non-negative values such that $a_1 + a_2 + \cdots + a_n = r$. For example, if we let $n = 3, r = 4$, we get (replacing x_1, x_2, x_3 with x, y, z)

$$(x + y + z)^4 = x^4 + \frac{4!}{3!1!} x^3 y + \frac{4!}{3!1!} x^3 z + \frac{4!}{3!1!} y^3 z + \cdots + \frac{4!}{2!2!} x^2 y^2 + \cdots.$$

6.2.9 Notice that there are four possibilities per person: no treat, just ice cream, just cookie, both treats. Then we have some correcting to do, since not all possibilities should be counted.

6.2.10 Where does the number 232 come from?

6.2.17 It is helpful to count the number of arrangements for which the sport-utility vehicle cannot park: next to each compact car, there will either be a single vacant space, or not.

6.2.25 It is easiest to assume that the trees are distinguishable; i.e., order matters. Thus there are $12!$ different ways of arranging the trees, and hence $12!$ will be the denominator of our probability. Then, when we count the number of arrangements for which no two birches are adjacent, we also assume that trees are distinguishable. This problem is not too different, then, from 6.2.17.

6.2.29 Look at the previous problem.

6.2.31 Look at a concrete example. Consider selecting 8 people from a pool of 20, of which 13 are men and 7 are women. What are the different cases?

6.2.34 The fact that $c(n, m)$ is a sum suggests a particular tactic. Which one? Show that if $A \subset B$, then $f(A) \leq f(B)$. If you know just a few crucial values of f , that will determine f completely.

6.3.10 It is helpful to compute the number of n -digit strings whose product is not a multiple of 10. Then the properties that matter are *not* being even, and *not* being equal to 5.

6.3.12 As with many PIE problems, it is helpful to count the complement. The permutations to focus on are not the ones that fix exactly k elements, but rather, the ones which fix element $\#k$. These will overlap, but that's what PIE is designed for.

6.3.15 Once again, count the complement: in how many ways can we give the cones out, and not use all k flavors? Focus on arrangements for which flavor $\#i$ is not used.

6.4.4 Either a word begins with a single-letter word, or it does not.

6.4.7 You may want to consider an auxiliary problem: how many legal sequences of n pairs of parentheses are "prime," in the sense that when you read from left to right, no substring is legal? For example, " $()()$ " is prime, but " $()()$ " is not.

6.4.22 (h) Adapt the same reasoning as in (b) to conclude that the expected number of c -cycles (loops of length c in a permutation) is equal to $1/c$, independent of the number of things being permuted.

Chapter 7

- 7.1.3** (e) Notice that (d) implies that if $g|a$ and $g|b$, then $g|a - b$.
- 7.1.11** Study Example 7.1.8 on page 236. Another approach is to use the Euclidean algorithm (Problem 7.1.12).
- 7.1.13** (b) Imagine that (a, b) and (u, v) are two different solutions to $17x + 11y = 1$. Show that $a - u$ must equal a multiple of 11, while $b - v$ is a multiple of 17. At some point you will use FTA.
- 7.1.17** Consider $ab/(a, b)$. Show that this must be a multiple of both a and b and thus be at least $[a, b]$. Now look at $ab/[a, b]$.
- 7.1.24** Look at the parity of the numerator and denominator of this sum (in lowest terms).
- 7.1.23** First try Problem 7.1.22.
- 7.1.26** Notice that a number of the form $4k + 3$ must have at least some prime factors of this form. Why?
- 7.1.32** What else do you know about consecutive numbers?
- 7.1.33** First show that $k|a_k$ for each k .
- 7.2.6** Look at perfect squares modulo 3.
- 7.2.15** The answer is “no.” Use Fermat’s little theorem.
- 7.3.19** Use the Gaussian pairing tool.
- 7.3.23** The left-hand side is a PIE statement.
- 7.4.7** Without loss of generality, assume $a \geq b \geq c$. Note that we must have $2 \leq (1 + 1/c)^3$. What does this tell you about c ?
- 7.4.10** The number 1,599 should force you to think hard about viewing the problem mod m , for a well-chosen m .
- 7.4.15** This already appeared as Problem 3.4.30 on page 107.
- 7.4.22** The pair 8, 9 is suggestive.
- 7.5.15** Use the methods of Problem 7.1.22 to count prime powers in numerator and denominator.
- 7.5.16** There are no non-zero solutions. Look at parity.
- 7.5.17** Two numbers are floating around: 3, and n , where $n \perp 3$. What do you do when you see relatively prime numbers?
- 7.5.24** Don’t forget the Chinese remainder theorem (7.2.18).
- 7.5.27** Let the number of cards be $2n$. Look at things modulo $2n + 1$.
- 7.5.35** Define $f(n)$ to be the sum of the zeros of $\Phi_n(x)$. Then, by 7.5.33, we see that $\sum_{d|n} f(d)$ equals the sum of the zeros of $x^n - 1$, and this is equal to 0, unless $n = 1$, in which case it equals 1. In other words. But by the MIF, we already know of a function with this property: μ . Recall that $\sum_{d|n} \mu(d)$ equals 0 unless $n = 1$, in which case it equals 1. So $f(n) = \mu(n)$.

Chapter 8

8.2.26 The diagonal of any quadrilateral dissects it into two triangles, and you already know things about triangles and midpoints.

8.2.27 Whenever you encounter a right triangle, remember that it inscribes nicely in a circle, with the hypotenuse as diameter. Always associate hypotenuses with diameters!

8.2.28 (g) The given point, tangent, and circle center make a right angle. Remember: right angles can be found inscribed in circles!

8.2.28 (k) There are two types of tangent lines. The “outside” ones, which intersect on one side of both circles; the “inside” ones, which intersect “between” them. Here is one case: Let the radii be R, r , with $R > r$. To construct an outside tangent line to both circles, you need to look at a right triangle, with one leg of length $R - r$ whose hypotenuse is the line segment connecting the two centers.

8.2.30 Let A, B, C denote the points of tangency between ℓ_1 and ω , ω and γ , γ and ℓ_2 , respectively. Let P, Q denote the centers of ω, γ , respectively. Note that P, B, Q are collinear, and that $AP \parallel QC$, and that triangles CQB and APB are isosceles. You may want to try an argument by contradiction: “Let A' denote the intersection of line CB with line AP . If $A' \neq A$, then ...”

8.2.31 A fold is a reflection across the fold line. Each point P is reflected to a point P' such that the fold line is the perpendicular bisector of PP' . These perpendicular bisectors give you lots of angle chasing opportunities.

8.2.33 You can easily determine which circle the midpoints must lie on, since you are told that they lie on a circle. To prove that they lie on this particular circle, you need the right penultimate step; use 8.2.13.

8.3.17 Look at the circumcircle of the medial triangle.

8.3.19 Compare areas.

8.3.21 (b) Extend the sides of the trapezoid upwards to form a triangle. (This is often a good idea when looking at trapezoids.) Now you have several similar triangles to contemplate that are each similar to this new triangle, and you also have two smaller similar triangles inside the trapezoid. Surely this is enough information!

8.3.27 Draw the angle bisector through A , which intersects BC at D' . Use the angle bisector theorem to prove that D and D' must be the same point.

8.3.29 Use a little algebra. Let k be the ratio of similitude between the two given triangles, so, for example $AB/DE = k$. Then let $r = AX/XB$.

8.3.33 It's easier to prove (d) before (c), since (d) implies (c). Just draw the appropriate diameter of the circle and look for an inscribed right triangle. Then you can read off the sine instantly. For (e), try (if C is obtuse) extending BC to D so that BDA is a right triangle with right angle at D . Now you have right triangles to read off trig functions with ease, and you can use the pythagorean theorem to get c^2 .

8.3.39 It is easier to look at the ratio $CE : EB$, which can be found with angle chasing and law of sines. You will need to use calculus, or the result of the previous problem. A very useful corollary of this is $\lim_{\theta \rightarrow \infty} (\sin a\theta / \sin \theta) = a$, for any constant a .

8.3.42 Midpoints suggest equal areas. You'll need auxiliary lines.

8.3.43 Draw a line tangent to the escribed circle that is parallel to BC .

8.4.7 You'd guess—correctly—that this ratio is constant, which means you have to hunt for similar triangles. Draw appropriate diameters to get right triangles, and look for the crux angle, one that is inscribed simultaneously in both circles. This angle will be the bridge that allows you to compare angles inscribed in one circle with angles inscribed in the other.

8.4.8 Perpendiculars dropped to sides are practically begging you to consider area. Do you know the formula for the area of an equilateral triangle in terms of its side length? If not, work it out now.

8.4.9 Think about the penultimate step for showing an angle to be constant. One such penultimate step would be for this angle to be inscribed in a circle, subtending a chord of constant length.

8.4.14 One penultimate step for this equality would be showing that a circle can be *inscribed* in quadrilateral $A'B'C'D'$. There are other approaches as well. And please notice that there are other cyclic quadrilaterals in this problem.

8.4.15 One way to show that these three points lie on a line is to show that $\triangle YJZ \sim ZIX$. Look for other similar triangles to prove this.

8.4.17 Make sure that you have studied Example 8.4.2. The answer, by the way, is $1/6$.

8.4.18 Parallel lines give birth to parallelograms; look for equal areas.

8.4.24 One way to show that XY is invariant is to show that it is a chord in a circle that subtends a constant angle. Look for a cyclic quadrilateral.

8.4.32 You may want to defer working on this problem until you have read about symmetry and reflections in the next section. The 60° angle and the perpendiculars creates many equal angles and equal sides...

8.4.33 Angle chase to find inscribed right angles.

8.4.34 The ratios in the problem suggest looking at similar triangles. Believe it or not, this problem succumbs to simple angle chasing, but you need to be *very* careful drawing the diagram. Color pencils are helpful to mark equal angles. Don't forget the tangent version of the inscribed angle theorem, and the fact that the measure of an exterior angle of a triangle is the sum of the opposite interior angles. Finally, you may want to compare the ratios AM/BM with EG/EF first.

8.5.18 Contemplate the parallelograms in this hexagon.

8.5.19 One method is to use vectors and efficient algebra. This is worth trying. Another, more “transformational” idea is the following: Suppose you are given the midpoints A', B', C', D', E' of a pentagon $ABCDE$, such that A' is the midpoint of AB , B' is the midpoint of BC , etc. Consider the rotation $R_{A', \pi}$. This takes A to B . Likewise, $R_{B', \pi}$ takes B to C . Now consider the composition of rotations

$$R_{E', \pi} \circ R_{D', \pi} \circ R_{C', \pi} \circ R_{B', \pi} \circ R_{A', \pi}$$

with the single rotation $R_{A', \pi}$. A translation is lurking about.

8.5.20 Notice that the height is fixed. If you need more hints, look at Example 3.1.5.

8.5.21 Vectors.

10 HINTS TO CHAPTER 9

8.5.23 To construct $M(T)$, try to build a triangle by translating the medians by vectors that are parallel to the sides of the original triangle. Vector notation may be helpful with the other parts of the problem.

8.5.25 Make sure that you understand the following: suppose that point X lies on line segment AB , with $AX : AB = r$. Then

$$\vec{X} = \vec{A} + r(\vec{B} - \vec{A}).$$

For **(a)**, consider the Euler line. For **(b)**, use the fact that the incenter is the intersection of the angle bisectors of the triangle. The method of “weights” (Problem 8.4.20) may also be useful.

8.5.30 Find rotations that take one of AX, BY, CZ to another. Since rotations preserve length, you’re done!

8.5.32 By now you probably realize that the penultimate step is a rotation. Symmetrical “points of interest” (besides the given points) are the midpoints of the sides of the parallelogram, and, of course, the center of the parallelogram.

8.5.36 The pantograph is nothing more (or less) than a homothety machine; you should have no difficulty in finding the center and scaling controls, etc.

8.5.48 Invert about E , showing that the image of the four reflection points lies on a rectangle.

Chapter 9

9.2.9 It turns out that $a_n \rightarrow \phi$, where $\phi := (1 + \sqrt{5})/2$. To see this, try drawing a picture (see Example 9.2.2 on page 329), or define the “error” sequence (e_n) by $e_n := a_n - \phi$, and then try to express e_n in terms of e_{n-1} with the goal of showing $e_n \rightarrow 0$. You will use, somewhere along the way, the fact that $\phi^2 - \phi - 1 = 0$.

9.2.11 Let $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2 + a_n}$. Since the (only positive) solution to $x = \sqrt{2 + x}$ is $x = 2$, you should try to prove that $a_n \rightarrow 2$.

9.2.17 Without loss of generality, try to get within 0.1 of an arbitrary point $x \in [0, 1]$. Divide $[0, 1]$ into 10 equal parts and use the pigeonhole principle. It doesn’t work immediately, but don’t give up!

9.2.23 Draw a careful graph. Make sure that you understand the graphical relationship between a function and its inverse function (the graph of $y = f^{-1}(x)$ is the reflection of the graph of $y = f(x)$ about the line $y = x$).

9.2.24 Think about $f^{-1}(500)$.

9.2.25 Define $g(x) := f(x + 1/1999) - f(x)$. It suffices to show that $g(x)$ changes sign for two values of x in the interval $[0, 1998/1999]$. If the endpoints don’t work out, be creative.

9.2.28 Is there anything special about $\sqrt{2}$? Is there a more general statement?

9.2.30 Pick $\epsilon > 0$. Then there is an N such that $|x_n - x_{n-1}| < \epsilon$ for all $n \geq N$. Now get a handle on the size of x_n for $n \geq N$ by telescoping, and using the triangle-inequality (see hint for 2.3.26).

9.2.31 Do Problem 9.2.30 first.

9.2.34 Look at Example 5.3.4 on page 168.

9.3.12 Rolles theorem.

9.3.16 Use the definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

9.3.26 Let $f(x) := 2x - 3x^3$. It's too hard to solve for the values $x = a, b$ at which $f(x) = c$, so just call these values a and b . Then our problem involves two integrals, one from a to b , and the other from 0 to a . Can these two integrals be combined?

9.3.29 Try to show that there is no such function. Can you find a polynomial $p(x)$ so that $\int_0^1 p(x)^2 f(x) dx = 0$?

9.3.32 Try a big-Oh simplification.

9.3.34 Take some time to figure out what the problem is asking. Your intuition tells you that it is plausible for $f(x)$ to be bigger than e^x . Might $f(x)$ exceed, say, e^{2x} for large enough x ? Logarithmic differentiation will help.

9.3.37 Look at the hint for 9.3.16 above. Also, recall that one way to prove that something equals a constant is to show that its derivative is zero.

9.3.38 The triangle inequality (see hint for 2.3.26) may be useful.

9.4.13 Use the generalized binomial theorem (9.4.12). Logarithmic differentiation may also be helpful.

9.4.17 Look at partial sums. There are geometric series to be summed.

9.4.19 What kind of function does $p_n(x)$ approach as $n \rightarrow \infty$?

9.4.23 Use the ideas of Example 9.4.7 on page 358.

9.4.29 The basic idea: look at n_k that have many prime factors.

9.4.32 Use PIE.